

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA0305	Course Title:	Natural Language Processing
CO Assessed:	CO3 – Precision	Assessment:	Assessment Tool 2 – Case Study
Weightage:	40% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
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Section / Batch:	2024 Batch	Date:	08/08/2026

Code of Conduct

I _____D.Chinna Moulali_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____D.Chinna Moulali_____

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3

CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus:

"Data science is powerful, data science drives innovation, data science is evolving."

The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences.

The following statistics are obtained from the corpus:

- $C(\text{data}) = 3$
- $C(\text{data science}) = 3$
- $C(\text{science is}) = 2$
- $C(\text{science drives}) = 1$
- $C(\text{science drives innovation}) = 1$

The interpolation weights are:

- $\lambda_1 = 0.5$ (Trigram)
- $\lambda_2 = 0.3$ (Bigram)
- $\lambda_3 = 0.2$ (Unigram)

The prediction probabilities after the word "science" are:

- $P(\text{is}) = 0.66$
- $P(\text{drives}) = 0.33$

Questions

1. Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability $P(\text{science} \mid \text{data})$. Interpret how this probability influences next-word prediction in a mobile keyboard application.
 2. A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.
 3. Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.
 4. Calculate the Entropy of the distribution $P(\text{is})=0.66$ and $P(\text{drives})=0.33$. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.
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CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.

Consider the following user inputs:

1. "Book a flight ticket now."
2. "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging.

Given:

- $P(\text{book} \mid \text{VB}) = 0.6$
- $P(\text{book} \mid \text{NN}) = 0.4$
- $P(\text{Start} \rightarrow \text{VB}) = 0.5$
- $P(\text{Start} \rightarrow \text{NN}) = 0.5$

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

Questions

1. Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.
 2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.
 3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.
 4. Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.
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CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.

The sentence processed by the system is:

"Economic growth increases employment."

Initial POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:

Change NNS to VBZ if the preceding word is tagged as NN.

The system also performs corpus-based frequency analysis and uncertainty evaluation.

Questions

1. Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.
2. Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.
3. Assuming the corpus contains the words:
 - economic (120 occurrences)
 - growth (450 occurrences)
 - increases (210 occurrences)
 - employment (380 occurrences)

Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.

4. Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.
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CASE STUDY 1: Smart Mobile Keyboard Prediction System

1. MLE of Bigram Probability $P(\text{science} \mid \text{data})$

The Maximum Likelihood Estimate for a bigram is:

Interpretation

The probability of predicting **"science"** after **"data"** is **100%** based on the given training corpus.

Therefore, when a user types **"data"**, the keyboard model has very high confidence that the next word is **"science"**.

In a real mobile keyboard, a high probability means that the word can be ranked near the top of the suggested next-word list.

2. Backoff Model for "data science improves"

The sequence **"data science improves"** is unseen because the corpus does not contain:

data science improves

Therefore, a trigram model cannot directly assign a reliable probability to the complete sequence.

A backoff model uses a lower-order model when the higher-order n-gram is unseen.

The available information is:

But **"improves"** has not been given any count in the corpus. Therefore, the bigram: cannot be estimated from the supplied statistics.

Thus, using the given corpus statistics, the sequence receives:

under an unsmoothed backoff model because **"improves" is completely unseen.**

Why is backoff essential?

Backoff is important in predictive keyboards because users constantly type new or rare word combinations. If a trigram is unavailable, the model can fall back to:

Trigram $\rightarrow$ Bigram $\rightarrow$ Unigram

This prevents the system from failing whenever an exact word sequence has not appeared in the training corpus.

For example:

data science improves

↓

Trigram unavailable

↓

Use lower-order bigram

↓

Use unigram if necessary

Therefore, backoff improves **coverage and robustness** in real-time prediction.

3. Deleted Interpolation for "data science is"

The interpolation formula is:

Given:

-
-
-

For the sequence:

data science is

From the corpus:

The trigram **data science is** occurs once in the corpus.

Also:

Therefore:

For the bigram:

For the unigram probability, the question does not provide the total corpus word count or . Therefore, the exact unigram probability cannot be calculated from the supplied information.

So:

If the intended simplified assumption is to use only the provided trigram and bigram values and treat the unigram contribution as unavailable, then:

Interpretation

Deleted interpolation combines:

- **Trigram:** more context but sparse
- **Bigram:** less context but more reliable
- **Unigram:** least context but broad coverage

The weights determine how much importance is given to each model.

Thus, interpolation provides a balance between **accuracy and coverage**, especially when training data is sparse.

4. Entropy of P(is) = 0.66 and P(drives) = 0.33

Entropy is:

Therefore:

Calculate:

Interpretation

The maximum entropy for two equally likely outcomes is:

The obtained entropy is approximately:

This indicates **moderate-to-high uncertainty** because both words are possible, although **"is" is more likely than "drives."**

The keyboard therefore has:

- Highest confidence in **"is"**
- Lower confidence in **"drives"**
- Some uncertainty because both predictions are possible

Role of entropy

Entropy helps a keyboard measure how uncertain its predictions are:

- **Low entropy** → **high prediction confidence**
- **High entropy** → **high uncertainty**

Therefore, entropy can help intelligent keyboards decide how many suggestions to display and how confidently to rank predictions.

CASE STUDY 2: AI-Powered Customer Support Chatbot

5. Penn Treebank POS Tags

Sentence 1

"Book a flight ticket now."

Word POS Tag Explanation

Book	VB	Verb
a	DT	Determiner
flight	NN	Noun
ticket	NN	Noun
now	RB	Adverb

Therefore:

Book/VB a/DT flight/NN ticket/NN now/RB

Here **book** is a **verb** because it expresses an action or command. The sentence is an imperative sentence.

For example:

Book a flight ticket.

Means **perform the action of booking**.

Sentence 2

“This book is interesting.”

Word	POS Tag	Explanation
This	DT	Determiner
book	NN	Noun
is	VBZ	Verb
interesting	JJ	Adjective

Therefore:

This/DT book/NN is/VBZ interesting/JJ

Here **book** is a noun because it refers to a physical or conceptual object.

Key Difference

Book a flight ticket.

↑

VB – action

This book is interesting.

↑

NN – object

The surrounding context determines the POS tag.

6. HMM likelihood of “book” being VB

Given:

and:

Using the simplified HMM calculation:

For NN:

Therefore:

Answer

The HMM favors **VB** because:

Hence, **book is more likely to be a verb** in the given context.

The contextual words **“a flight ticket now”** further support the interpretation that “Book” is an action.

7. Rule-Based Tagging vs Stochastic HMM Tagging

Feature	Rule-Based Tagging	HMM/Stochastic Tagging
Method	Hand-written linguistic rules	Probabilities learned from corpus
Ambiguity	Depends on manually designed rules	Handles ambiguity probabilistically
Training	Usually no statistical training	Requires training data
Adaptability	Lower	Higher
Large datasets	Difficult to maintain	More suitable
Context	Uses predefined rules	Uses probabilities and context
Scalability	Limited	Better

Rule-Based Tagging

A rule might state:

If “book” occurs before a determiner/object → VB

Such rules work for known patterns but can become complicated as the vocabulary and contexts increase.

HMM Tagging

An HMM calculates probabilities such as:
and transition probabilities between tags.

It can therefore select the most probable tag based on statistical evidence.

Recommendation

For a **large-scale chatbot**, stochastic/HMM-based tagging is generally more suitable because it:

- Handles lexical ambiguity
- Uses contextual information
- Can learn from large corpora
- Scales better to large datasets
- Can adapt to different language patterns

A practical modern system would generally use more advanced statistical/neural NLP models, but among the two approaches given here, **HMM tagging is preferable**.

8. Role of Standardized POS Tagsets

A standardized POS tagset such as the **Penn Treebank tagset** provides consistent labels for grammatical categories.

Examples:

NN → Noun

VB → Verb

JJ → Adjective

RB → Adverb

DT → Determiner

Importance in chatbot systems

9. Intent detection

POS information helps distinguish actions from objects.

For example:

Book a flight

Here **book/VB** indicates an action, helping identify a booking intent.

10. Response generation

The chatbot can identify grammatical structure and generate more appropriate responses.

11. Word ambiguity

The same word can have different grammatical roles:

Book a flight → book/VB

This book is good → book/NN

POS tagging helps resolve this ambiguity.

12. Better NLP processing

Standardized tags provide a common representation for downstream NLP components.

Conclusion

Standardized POS tagging improves:

- Intent classification
- Lexical ambiguity resolution
- Information extraction
- Response generation
- Overall language understanding

Therefore, POS tagging is an important preprocessing component in intelligent chatbot systems.

CASE STUDY 3: News Analytics and POS Tag Correction System

1. Apply the Transformation Rule

Initial tags:

economic/JJ

growth/NN

increases/NNS

employment/NN

Rule:

Change NNS to VBZ if the preceding word is tagged as NN.

The word **increases** is initially tagged as NNS.

Its preceding word is:

growth/NN

Therefore, the rule applies.

Updated POS sequence

economic/JJ

growth/NN

increases/VBZ

employment/NN

Explanation

The sentence is:

Economic growth increases employment.

Here:

- **economic** → adjective modifying growth
- **growth** → singular noun and subject
- **increases** → singular present-tense verb
- **employment** → noun functioning as object

The subject **growth** is singular, so the verb should be **increases**, which takes the **VBZ** tag.

Therefore, the transformation is linguistically appropriate.

2. Analyze Initial Tagging Errors

The initial tagging was:

economic/JJ

growth/NN

increases/NNS

employment/NN

The main error is:

increases/NNS

Why is NNS incorrect?

NNS means a plural noun.

Examples:

students/NNS

books/NNS

cars/NNS

But **increases** in this sentence is not a plural noun. It is a **third-person singular present-tense verb**.

Therefore:

increases/VBZ

is the correct tag.

Grammatical structure

The sentence can be represented as:

Subject	Verb	Object
↓	↓	↓
economic growth	increases	employment

More specifically:

economic/JJ

↓

growth/NN

↓

increases/VBZ

↓

employment/NN

The singular subject **growth** agrees with the singular verb **increases**.

3. Word Frequency Distribution

Given:

Word	Frequency
economic	120
growth	450
increases	210
employment	380

Step 1: Calculate total frequency

Total:

Step 2: Calculate probability/frequency distribution

The probability of each word is:

economic

growth

increases

employment

Frequency Distribution Table

Word	Count	Probability	Percentage
economic	120	0.1034	10.34%
growth	450	0.3879	38.79%
increases	210	0.1810	18.10%
employment	380	0.3276	32.76%
Total	1160	1.0000	100%

Role in probabilistic POS tagging

Frequency information helps a probabilistic tagger estimate how likely a word or tag is.

For example, if a word frequently occurs as a verb in the training corpus, the model can assign a higher probability to its verb tag.

However, **word frequency alone is not sufficient** to determine POS. Context and neighboring tags are also important.

Thus, frequency information provides statistical evidence for POS tagging.

4. Transformation-Based Tagging and Entropy

How Transformation-Based Tagging improves accuracy

Transformation-Based Tagging, such as the **Brill Tagger**, typically works in two stages:

Input sentence



Initial POS tagging



Find tagging errors



Apply learned transformation rules



Corrected POS tags

In this example:

Before transformation

economic/JJ

growth/NN

increases/NNS

employment/NN

Transformation

NNS → VBZ

if preceding word = NN

After transformation

economic/JJ

growth/NN

increases/VBZ

employment/NN

Therefore, the transformation corrects the tagging error without requiring the entire sentence to be tagged again.

Advantages

Transformation-based tagging:

- Corrects systematic errors
- Uses contextual information
- Improves initial tagging accuracy
- Learns useful linguistic patterns
- Is relatively interpretable because rules can be examined

Entropy for Measuring Uncertainty

Entropy is:

where represents possible POS tags.

Suppose the word **increases** initially has two possible tags:

NNS = 0.5

VBZ = 0.5

Then:

This indicates high uncertainty.

After applying the transformation rule, suppose the model strongly favors VBZ:

$P(\text{VBZ}) = 0.9$

$P(\text{NNS}) = 0.1$

Then:

So:

The decrease in entropy indicates that the system has become **more confident** about the POS tag.

Overall Conclusion

Transformation-Based Tagging improves POS tagging by applying contextual correction rules to initial predictions. Entropy provides a quantitative measure of uncertainty:

- **High entropy → uncertain POS assignment**
- **Low entropy → confident POS assignment**

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____